Model Comparison Table

Validation Set Results

Classification Model	Accuracy	Precision	Recall	F1 Score	ROC/AUC
Logistic Regression	0.930	0.408	0.891	0.560	0.977
Decision Tree Classifier	0.944	0.458	0.594	0.517	0.778
Random Forest Classifier	0.965	0.631	0.745	0.683	0.977
Gradient Boosting Classifier	0.939	0.442	0.855	0.583	0.976
K-Nearest Neighbors Classifier	0.931	0.409	0.842	0.550	0.942
Support Vector Classifier (SVC)	0.954	0.521	0.903	0.661	0.985
Voting Vs Average (best 3 of 6)	0.963	0.594	0.806	0.684	0.982
Ensemble (Bayesian Weighted)	0.963	0.594	0.806	0.684	0.982

Test Set Results

Classification Model	Accuracy	Precision	Recall	F1 Score	ROC/AUC
Logistic Regression	0.936	0.433	0.945	0.594	0.977
Decision Tree Classifier	0.951	0.505	0.677	0.578	0.821
Random Forest Classifier	0.966	0.632	0.774	0.696	0.980
Gradient Boosting Classifier	0.936	0.430	0.878	0.577	0.980
K-Nearest Neighbors Classifier	0.920	0.368	0.841	0.512	0.936
Support Vector Classifier (SVC)	0.951	0.505	0.927	0.654	0.986
Voting Vs Average (best 3 of 6)	0.961	0.573	0.884	0.695	0.985
Ensemble (Bayesian Weighted)	0.961	0.574	0.878	0.694	0.985

Which model gave me the best result, and which gave the worst?

The best was the Random Forest Classifier which produced the strongest standalone performance, with the highest F1 Score on both validation 0.683 and test 0.696 and the best precision of any individual model 0.632 validation, 0.632 test. It offered the most balanced tradeoff between catching real heatwaves and avoiding false alarms.

The worst one was the K-Nearest Neighbors Classifier produced the weakest results, with the lowest F1 Score on both validation 0.550 and test 0.512 and the lowest overall accuracy. Its precision was especially bad at 0.37 - 0.41, which means a large share of its predicted heatwaves were false alarms.

One thing to note on the Ensembles is the Voting and Bayesian ensembles outperformed every individual model slightly on ROC-AUC and recall, but at the cost of somewhat lowering the precision compared to the Random Forest alone. Which one is preferable is up to whether missed heatwaves or false alarms are more costly in a real-world deployment, which I talked about in the analysis report.